REFERENCE · GENALPHA-BSS

The Operator's Manual

How to run, configure and operate the BSS — by role and by task. The companion volume, The Honest Machine (a build memoir · verify everything — how one person and an AI built a complete telecom suite and proved every piece of it), tells how it was built and why; this book tells you which button, which endpoint, which environment variable. Everything here is proven by the suite named beside it.

Contents

1 · Getting started

```
docker compose up -d          # ~90 containers; first build takes a while
open http://localhost:8080/shop/   # the storefront (guest browsing works)
```

The gateway is `:8080`; Keycloak is `:8085` (admin/admin). Two seed operators run side by side: **genalpha** (EN/EUR, realm `bss`) and **nova** (NO/NOK, realm `nova`) — plus any operator you mint (§7). Demo card `4242 4242 4242 4242` pays; anything ending `0002` declines. Promo code `WELCOME10`. Serviceable fiber postcodes start `111` / `222` / `333`.

> **The one discipline:** every feature in this manual has an end-to-end Playwright suite in `ops/e2e/`. When in doubt about how something behaves, the suite is the specification: `node ops/e2e/<name>_test.js` runs it against the live stack.

2 · Surfaces & sign-ins

URL	WHO	DEMO SIGN-IN
`/shop/`	Customers (self-register or guest)	`kai@bss.local` / `kai`
`/app/`	Customers on mobile (Expo web)	`emil@acme.example` / `emil`
`/biz/`	B2B org admins + invited members	`bianca@acme.example` / `bianca`
`/csr/`	Agents (role-scoped powers)	`agent-anna` / `agent` ; junior: `jo@bss.local` / `jo`
`/console/`	Back office (role-scoped tabs)	`demo` / `demo` ; product-only: `pat@bss.local` / `pat`
`/dealer-app/`	Retail-partner clerks & telesales agents	any account whose org holds a dealer agreement
`shop.nova.localhost:8080/shop/`	Nova's white-label storefront	`nils@nova.local` / `nils`
`biz.nova.localhost:8080/biz/`	Nova's B2B console (Norwegian)	`birgit@fjellheim.no` / `birgit`

White-labeling is hostname-driven: `shop.<tenant>.localhost` serves that operator's brand, locale and currency from the live registry (§7).

3 · The product owner

Catalog (console → Product Offerings / Specifications / Prices)

Offerings are TMF620 data: hard and soft bundles (`bundledProductOfferingOption` cardinality, pick-N-of-M choice groups), characteristic-conditioned prices ("+2.00/mo when colour = Titanium"), commitment terms, categories that drive placement *and* fulfilment (Top-ups and Insurance are billing-only; Partner services mint entitlement codes; everything else draws an MSISDN + SIM). Artwork rides the offering's attachments — internal store or the tenant's own PIM.

Rules & pricing as data (console → Rules)

Order rules and dynamic pricing are JSON-logic rows with a dry-run panel. A price change is a rule, not a release. See `docs/product-rules.md` for the authoring guide.

Product Copilot (console → Copilot)

Chat a product into the catalog: the model proposes specs/prices/offerings as TMF620 payloads, the console shows a human card, and a deterministic executor applies it with *your* token on confirmation. The model never writes.

Product Advisor (console → Product advisor) — suite #52

Findings with receipts: top-up attach counted from inventory ("2 of 2 customers on this plan also bought top-ups"), market price gaps from the tenant's market feed (10% materiality threshold — small gaps are deliberately silent), an LLM narrative that never invents a number. **Adopt as draft…** births an `In study` offering; nothing reaches the shop until you promote it.

Campaigns & growth (console → Campaigns / Journeys / Audiences)

Segments form from consented shop behavior; campaigns prove lift against real holdouts; journeys read the customer before speaking; A/B arms split deterministically; guardrails (frequency caps, quiet hours) are settings. Every message lands on the TMF683 timeline.

4 · The CSR

Search the 360 (multi-word, name-ranked), then act — every action logs itself on the customer's timeline:

THE CUSTOMER ASKS	THE BUTTON / ACT
"What's my PUK?" / "reset my PIN"	Reveal PUK · Reset PIN (owner notified — a silent credential change reads as fraud)
"I lost my SIM"	Replace SIM (old card quarantined first)
"new number, please"	Change number (old number quarantined with its story)
"pause my subscription"	Pause / Resume (charging pauses at the OCS; auto-resumes on the promised date)
"I can't pay all of it"	Installments (parts sum exactly; miss one → reminder → grace → dunning)
"this charge is wrong"	Dispute (collection pauses; resolution credits or upholds, in writing)
"send me my invoice"	PDF (the exact document) · Email bill (to the address on file, never one dictated on the phone)
"my internet is slow"	Diagnose (triage across usage/assurance/network facts)
"give my number to my colleague"	Transfer (three-way authorization; B2B handover keeps the company paying)

Self-care mirrors most of these in the shop and app — customers act on their OWN line (ownership-checked; a foreign line answers 404, never 403).

5 · Billing & money operations

The billing run

`POST /tmf-api/customerBillManagement/v4/billingRun` (staff). Segment-based and per-account: mid-month starts, mid-cycle plan changes and ceases each bill exactly their own days, labelled ("from the 16th, 15 days"). Payday alignment per customer (`POST /individual/{id}/billingCycle` , day 1–28). The invariant the suites enforce from every direction: **no day is ever billed twice**.

The bill as a document — suite #46

`GET /customerBill/{id}/document.pdf` (owner-scoped); `POST /customerBill/{id}/resend` emails it to the address on file. Delivery preference per customer: `paper | einvoice | digital` (`POST /individual/{id}/billDelivery`) — the customer picks the channel; the tenant keeps the plumbing.

E-invoice formats are rows (console → Bill formats)

CODE	STANDARD	SYNTAX
`ehf`	EHF Billing 3.0 (Norway; carries the KID)	UBL 2.1
`peppol`	Peppol BIS Billing 3.0	UBL 2.1
`cii`	EN 16931	UN/CEFACT CII
`aunz`	A-NZ Peppol BIS 3.0	UBL 2.1
`edifact`	UN/EDIFACT INVOIC D96A	segments
`facturx`	Factur-X / ZUGFeRD	PDF + embedded CII

Adding a country is an INSERT (the suite added XRechnung through the form). The tenant's `BILL_DISTRIBUTION_FORMAT` picks a row by code; the renderer follows the row.

Delivery, status, and the money coming home — suites #46/#47

Deliveries (console tab): the outbox-backed ledger — every bill's trip, attempts, the buyer's Peppol Invoice Response (accepted / rejected with their words / paid), one-click Retry on failures. **Remittance**: the bank posts camt.054, Nets OCR or BAI2 to `POST /bank/v1/remittance` (per-tenant `BANK_TOKEN`); a matched KID settles the bill through the card path's own guarantee; everything unclear parks on **Unapplied cash** (console tab) with its reason. Money is fail-closed: never guessed, never dropped.

Dunning (console → Dunning)

The collections worklist: overdue installments, broken plans, what is still owed.

6 · Partner channels — suites #48/#51

> **Being a dealer is a row.** Sign a chain: `POST /dealer/v1/agreement` (staff) with the org, commission per activation, and — for a chain with its own POS — the OAuth2 `clientId` that speaks for them. Clerks are org membership; foreign dealers see 404-shaped nothing.

Retail (the CSP + external-retail model — think Elkjøp/Power)

Counter sale: sell against the customer's email; the order carries the dealer stamp. **Starter kits**: mint batches (code + SIM + attribution in the box); the customer self-activates (`POST`

`/dealer/v1/starterKit/activate`) and the line rides the SIM from the box. **POS API**: same endpoints, machine credential, per-partner rate limits at the edge (429 + Retry-After); the chain's own phone rides the sale as `device` context — never a billable item. **Commission**: pending → earned after the angrerett window → clawed back with the reason if the customer leaves inside it.

Telesales (outbound, shaped by the law)

Dial list: `GET /dealer/v1/telesales/dialList?segment=` — consented at the source (insight segments), every number DNC-washed, reserved citizens excluded *and counted*. **The call makes an OFFER** (`POST /dealer/v1/telesales/offer`): washed fail-closed (reserved refuses, unreachable register refuses, unconfigured tenant refuses); no order exists yet. **The customer confirms in writing** (`POST /telesales/v1/confirm`, signed in, code from their inbox — or, for a cold prospect, after registering with the offered email); only then the order, the activation and the commission are born. Silence expires on a clock. Every dialer call logs to TMF683.

7 · The host administrator

The tenant fleet is a file

`infra/tenants/tenants.yml` is the shared registry (mounted into all 31 services); every service live-refreshes it. NEW operators join the running fleet within one refresh interval; changes to a serving operator's seams/brand/hosts apply live; identity fields (`id`, `issuer`, key endpoints) require a restart — by design.

Minting an operator — suites #49/#50

```
# the script (~2 minutes to a billed customer):
ops/onboard-tenant.sh fjord "Fjord Mobil" da DKK "#1B6CA8"

# or the form: console → Operators → five fields → Save
#   (host-tenant admins only; realm clone + registry block + seeded catalog;
#    shop.<id>.localhost wears the brand the moment the gateway refreshes)
# live rebrand of a form-born operator:
PATCH /onboarding/v1/operator/{id}   {"name","color","locale","currency"}
```

Seed operators (genalpha, nova) are protected from the form — their config is env.

Staff & roles (console → Staff)

Grant/revoke whole areas per operator over the tenant's own IdP (TMF672) — no Keycloak console needed for daily administration.

8 · AI configuration — suite #53

Per tenant, all data: `ai-provider` (`stub` — keyless CI default — `openai-compatible` , `anthropic`), `ai-base-url` , `ai-api-key` , `ai-model` . **Tiered routing**: the task class lives at the call site — FAST (copywriting, summaries, narratives) vs SMART (product proposals, next-best-offer; the default when unclassified). Every value resolves tier-first:

```
AI_MODEL_FAST=swift-mini            # cheap model for volume work
AI_MODEL_SMART=frontier-x           # careful model for judgment
AI_PROVIDER_SMART=anthropic         # tiers go down to the PROVIDER:
AI_BASE_URL_SMART=https://api...    # a local endpoint for volume,
AI_API_KEY_SMART=...                # a frontier API for judgment — at once
```

9 · The seam catalog

Every external system is a per-tenant seam: configure it and the feature lights up; leave it blank and the feature degrades honestly (fail-open for enrichment, fail-closed for money, identity and consent).

SEAM	TENANT FIELDS (ENV)	BEHAVIOR WHEN BLANK
Email provider (ESP)	`DELIVERY_PROVIDER, ESP_URL, ESP_API_KEY, ESP_FROM`	in-app inbox only
Bill distribution partner	`BILL_DISTRIBUTION_{PROVIDER,URL,TOKEN,FORMAT,CHANNEL}`	bills stay in-app/PDF
Bank (remittance)	`BANK_TOKEN`	webhook refuses (money: fail-closed)
Do-not-call register	`DNC_URL, DNC_TOKEN`	outbound telesales refuses (opt-IN by configuring)
Market-intelligence feed	`MARKET_FEED_URL, MARKET_FEED_TOKEN`	advisor's market findings absent; own-data findings stand
LLM	`AI_*` (§8)	keyless stub answers
PIM (product imagery)	`PIM_BASE_URL`	internal TMF667 store
Web analytics (GA4)	`ANALYTICS_*`	internal insight only
Social platform	`SOCIAL_API_URL, SOCIAL_ACCESS_TOKEN`	no audience push / lead import

Per-tenant variants ride suffixes: `_NOVA`, `_FJORD`, ... matching the placeholders in `tenants.yml`.

10 · Environment reference (dev clocks)

VARIABLE	MEANING	DEV / PROD DEFAULT
`BSS_BILLING_DUNNING_TICK_MS / _GRACE_MS`	collections sweep / grace	5s & 20s / 60s & 7d
`BSS_BILLING_DISTRIBUTION_TICK_MS / _RETRY_SECONDS`	delivery relay / backoff base	3s & 4s / 30s & 60s
`BSS_SOM_COMMISSION_WINDOW_MS / _TICK_MS`	angrerett window / harden tick	15s & 5s / 14d & 1h
`BSS_SOM_TELESALES_EXPIRY_MS / _TICK_MS`	offer expiry / sweep	15s & 5s / 48h & 1h
`BSS_TENANTS_REFRESH_MS`	live registry refresh	15s
`BSS_GATEWAY_PARTNER_RATE_CAPACITY / _WINDOW_MS`	per-partner rate limit	10 per 2s / 60 per 60s

11 · Extension API reference (beyond the TMF standard surfaces)

ENDPOINT	WHO	DOES
`GET /customerBill/{id}/document.pdf`	owner/staff	the bill as a PDF
`POST /customerBill/{id}/resend`	owner/staff	email the invoice to the address on file
`POST /customerBill/{id}/dispute · /installmentPlan`	owner/agent	contest / split a bill
`GET/PATCH/POST /billFormatProfile[/{code}]`	read: billing · write: billing:admin	e-invoice profiles as rows
`GET /billDistribution · POST /billDistribution/{id}/retry`	billing:admin	the delivery ledger
`POST /bank/v1/remittance`	the bank (X-Bank-Token)	camt.054 / OCR / BAI2 in
`POST /distribution/v1/response`	the partner (X-Distribution-Token)	Peppol Invoice Response onto the ledger
`GET /remittance/unapplied`	billing:admin	unapplied cash worklist
`POST /individual/{id}/billingCycle · /billDelivery`	self/staff	payday alignment · delivery preference
`/dealer/v1/*` (agreement, kits, sell, commission, orders/{id}, telesales/*)	staff / dealer / partner POS	the whole dealer & telesales channel
`POST /telesales/v1/confirm`	the customer, signed in	the written yes that births the order
`GET /advisor/v1/findings · POST /advisor/v1/adopt`	catalog:write	product advisor · adopt-as-draft
`GET/POST /onboarding/v1/operator · PATCH …/{id}`	host roles:admin	mint / live-mutate an operator
`GET /app/tenant-config.json`	public, by Host	the white-label manifest

12 · The mock integrations (dev stand-ins)

CONTAINER	PORT	STANDS IN FOR	CHAOS / INSPECTION
mock-esp	8121	SendGrid-shaped email provider	`GET /mails`, webhook receipts
mock-social	8122	Meta-shaped audiences/leads	—
mock-distribution	8124	Peppol access point / print house (validates EHF/EDIFACT/Factur-X!)	`GET /invoices`, `POST /outage`, `POST /invoices/{billNo}/respond`
mock-dnc	8125	reservation register	numbers ending 9999 are reserved; `POST /outage`
mock-market	8126	tariff-comparison feed	`GET /offers`
mock-llm	8127	OpenAI- and Anthropic-dialect LLM	names the model in answers; `GET /requests`

genalpha-bss · fifty-three suites, eleven CTKs at zero failures · the build story is *The Honest Machine* (docs/book) · the TMF vocabulary is docs/book/tmf-reference.md